

Mohammed Moniruzzaman Khan

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Professional Summary

Quantitative Analyst & Data Scientist with expertise in machine learning, deep learning, and financial modeling. Experienced in pricing and hedging financial derivatives, developing AI-driven predictive models, and deploying Python-based solutions. Skilled at translating advanced mathematics and stochastic modeling into actionable, data-driven insights.

Education

Nova School of Science & Technology, Lisbon 2024–Present

PhD in Mathematics

Thesis: Future Research Problems: Pricing and Hedging Options with Machine Learning

Key Courses: Computational Financial Mathematics, Stochastic Differential Equations, PDE Analysis

Uppsala University, Sweden 2020–2024

Master of Science in Mathematics

Thesis: Pricing and Hedging American-Style Options with Deep Learning: Algorithmic Implementation

Supervisor: Prof. Kaj Nyström

Thesis Link: [Uppsala University Repository](#)

Key Courses: Data Science & Machine Learning, Probability Theory, PDEs, Functional Analysis

Baruch College, CUNY, USA 1993–1999

BA in Mathematics

Independent Study: Implementation of Data Structures Using Java, Prof. B. Shaw (Retired)

Core Competencies

- Python, MATLAB, SQL, Java, C
- TensorFlow, Keras, Scikit-Learn, Deep Learning, Feature Engineering, Time Series Forecasting
- PDEs, Stochastic Processes, Monte Carlo Simulations, Option Pricing
- Pandas, NumPy, Matplotlib, Seaborn, EDA, Statistical Testing
- ETL, Spark, Docker, AWS (basic)
- Predictive Analytics, Model Optimization, Risk Management

Projects & Certifications

- Customer Retention MLOps Pipeline: [GitHub](#) – Scalable pipeline predicting customer retention; deployed with Docker and AWS.
- Financial Time Series Forecasting: [GitHub](#) – Forecasted SPY, QQQ, BTC using ARIMA, Random Forest, LSTM; feature engineering included moving averages and rolling volatility.
- American Options Pricing with Deep Learning: [GitHub](#) – Implemented Longstaff-Schwartz Monte Carlo in MATLAB; neural networks enhanced hedging strategies.
- Computational Finance PDE Solver: [GitHub](#) – Python-based solvers for European/American options with Monte Carlo Black-Scholes modeling.
- Titanic Survival Prediction: [GitHub](#) – Random Forest ML pipeline with cross-validated accuracy 82
- Certifications: Kaggle – Intro to ML, Intro Advanced SQL, Pandas, Feature Engineering, Time Series Forecasting, Intro to Deep Learning; Le Wagon – Introduction to Docker

Professional Experience

The Stylex Collection, Dhaka, Bangladesh Sep 2018–Jan 2019

IT In Charge – Managed IT infrastructure, Windows Server, network configuration, and security policies; improved system reliability and productivity.

IFIC Bank Ltd, Dhaka, Bangladesh Jul 2013–Jul 2018
Senior Principal Officer – Administered SWIFT servers and customized Java-based MMS integrated with core banking; developed SQL Server components ensuring transactional data integrity.

IBCS-Primax Software Ltd, Dhaka, Bangladesh Nov 2012–Jun 2013

Senior Programmer – Designed and optimized Java enterprise software modules; created Oracle 10g database schemas.

UTC Associates (Pvt) Ltd, Dhaka, Bangladesh Jan 2012–Oct 2012

Software Engineer – Developed BI reports and optimized SQL queries for enterprise reporting.

ADN Technologies Ltd, Dhaka, Bangladesh Apr 2011–Nov 2011

UI Architect / Project Lead – Modernized legacy POS system in C; integrated with back-office modules.

Empire Art Gallery, NY, USA 1995–2009

UI Architect / Software Consultant – Developed web/internal applications using HTML, CSS, JavaScript, SQL; improved inventory and e-commerce systems.

Achievements

- Designed and implemented ML and deep learning models for pricing and hedging American-style options, demonstrating reproducible, high-accuracy workflows.
- Developed numerical PDE solvers and Monte Carlo frameworks for option pricing with optimized computational efficiency.
- Translated advanced mathematical theory (PDEs, SDEs, stochastic processes) into Python and MATLAB implementations suitable for research and industry applications.
- Built and maintained a [GitHub](#) portfolio showcasing end-to-end quantitative finance, machine learning, and data science projects.

Languages

English: Fluent

Bengali: Native

References

- Marta Faias: PhD Supervisor – mcm@fct.unl.pt
- Pedro Mota: PhD Supervisor – pjpm@fct.unl.pt